

STUDY HALL · FRIDAY 10:30

Week 1 Review Sheet

The 27 big ideas of Week 1 — three from each course — plus four NICU Tutor flashbacks. Read each one; if you can explain it out loud in your own words, check the box. Anything unchecked is what to reread before the final.

Freshman Seminar

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Lectures teach. Recitations practice (homework happens there). Exams prove.

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Notes are for future-me: 3 big ideas, in my own words.

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Quiz strategy: read twice, read all choices, cross out wrong ones, then choose.

Chemistry 101

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Everything is made of atoms. About 118 kinds (elements); living things use mostly C, H, O, N.

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A formula is a molecule's recipe. The subscript counts the atom just before it. No number = 1.

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A breath is a trade across the alveoli: O_2 in to the blood, CO_2 out. Too much CO_2 makes blood acidic.

Biology 101

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A cell is the smallest living thing. Ladder: atoms → molecules → cells → tissues → organs → systems → organism.

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Mitochondria are the cell's power plants: sugar + oxygen → energy, with CO_2 and water as exhaust.

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Cells divide by doubling (1 → 2 → 4 → 8...). Growth costs fuel — that's why preemies need so much.

Physics 101

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Pressure = push ÷ area. Same push on a smaller area = bigger pressure.

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Air has weight: at sea level it holds up 760 mm of mercury — so pressure (and blood pressure) is measured in mmHg.

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Breathing machines use gentle pressures in cmH_2O . 1 mmHg ~ 1.36 cmH_2O because mercury is 13.6× heavier than water.

Calculus 101

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Calculus is the math of change. Big question 1: how fast is it changing?

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Rate of change = change ÷ time. $(970 - 900) ÷ 7 \text{ days} = 10 \text{ grams per day}$. 'Per' means 'divided by'.

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On a graph, steepness = rate. Steep = fast, flat = none, downhill = losing (negative). Read the line, not one dot.

Statistics 101

- Mean = add them all, divide by how many. Range = biggest - smallest.
- Apgar score (1952): 5 signs $\times$ 0-2 points = 0-10 at 1 and 5 minutes. 7-10 good; 4-6 may need help; 0-3 needs help now.
- Numbers can be compared and studied in thousands; 'looks fine' can't. But the mean hides the spread – ask for the range.

Genetics 101

- DNA = the instruction book in 4 letters (A, T, C, G), 46 chromosomes in 23 pairs – one of each pair from each parent. A gene = one recipe for one protein.
- Cells read DNA 3 letters at a time ($4 \times 4 \times 4 = 64$ words) to build proteins. Every cell has the whole book; each type reads different chapters.
- Identical twins = one egg split (same DNA). Fraternal = two eggs. Twins are usually born early and small – so they fill NICU beds.

Psychology 101

- Babies can't talk, so psychologists measure what they DO: looking time, sucking, heart rate.
- Newborns already hear (from ~24 weeks), recognize mom's voice, prefer faces, and see best at ~20-30 cm.
- A newborn brain wires at over a million connections a second, shaped by experience. Premies wire in the NICU – so: dim, quiet, protected sleep, parents close.

Sociology 101

- Sociology = people in groups: roles, hierarchy, rules, rituals. A NICU is a small society.
- The NICU team: doctors, nurses (most bedside hours), respiratory therapists, pharmacists, therapists, social workers, cleaners, parents. Nobody does it alone.
- Levels I-IV; regionalization gets the sickest babies to the right level. Rounds are a ritual that makes the team reliable. Good teams let anyone speak up.

NICU Tutor flashbacks

- Oxygen has a dose: premies on oxygen are kept around 90-95% because too much harms eyes and lungs.
- Surfactant coats the alveoli so they don't stick shut; premies don't make enough (RDS).
- Jaundice: bilirubin from recycled red blood cells builds up faster than a new liver can clear it; blue light fixes it.
- The fontanelle is a bone-free window for brain ultrasounds; IVH is the premie brain bleed doctors screen for.